

Influence_Consensus_agr_inn - In society, at the population level, does Agriculture alone generally have a direct effect on Innovation, or are observed relationships primarily indirect? This is a macro level analysis, including all of society, not just agrarian or farming populations. If a direct effect exists, is it negative, none, negligible, or positive? What would be a reasonable normalized magnitude for this influence on a scale from -1.00 to 1.00? Please only provide evidence for only direct influences

Agriculture and Innovation: Evidence for Direct Macro-Level Effects

Agriculture is typically treated in the literature as a **recipient and generator of innovation**, not as an exogenous driver of overall societal innovation. The available papers focus on how innovation occurs within agricultural systems and how policies, networks, and technologies shape agricultural innovation, rather than quantifying agriculture's direct effect on innovation across the whole economy.

Nature of the Agriculture–Innovation Relationship

Agriculture as an innovation system, not a macro driver

- Several reviews conceptualize **agricultural innovation systems** (R&D, training, extension, ICT, sustainability) as subsystems within broader national innovation systems, stressing internal interactions among farmers, research, policy, and markets (Rosario et al., 2022; Ogundari & Bolarinwa, 2018).
- Innovation in agriculture is described as occurring through **technology adoption, social innovation, and institutional arrangements** within rural/agri-food contexts, not as a measured cause of innovation in other sectors (Cano et al., 2023; Da Silva et al., 2024; Elsner et al., 2023; Gouroubera et al., 2025; Glover et al., 2019).

Evidence on Direct Effects (or Lack Thereof)

- Studies and reviews analyze:
 - Impacts of **agricultural innovations on yields, income, social outcomes** (Basso et al., 2024; Campuzano et al., 2023).
 - Determinants of **technology adoption in agriculture** (behavioral factors, policy, networks) (Pigatto & Brunori, 2021; Nakano et al., 2018; Li et al., 2024; Schulz & Börner, 2022; Cano et al., 2025; Bakali et al., 2023).
 - **Social innovation** in agricultural and rural systems (short food chains, value-added products, territorial development) (Elsner et al., 2023; Gouroubera et al., 2025; Glover et al., 2019).
- None of these quantify agriculture's share in **overall societal innovation output** (e.g., total patents, cross-sector R&D) or estimate an economy-wide elasticity of "agriculture → innovation".

Focus of Key Papers

Main focus of papers	Relation to "agriculture → innovation (macro)"	Citations
Scaling & systems of agri innovation	Internal to agriculture only	(Cano et al., 2023; Rosario et al., 2022; Ogundari & Bolarinwa, 2018; Bakali et al., 2023)

Main focus of papers	Relation to “agriculture → innovation (macro)”	Citations
Adoption & impact of agri technologies	Innovation within farms/sectors	(Basso et al., 2024; Pigatto & Brunori, 2021; Nakano et al., 2018; Li et al., 2024; Schulz & Börner, 2022; Cano et al., 2025; Campuzano et al., 2023)
Social innovation in agri/rural systems	Rural development, not macro innovation	(Elsner et al., 2023; Gouroubera et al., 2025; Glover et al., 2019)
Patent trends in agri tech	Innovation in agriculture, not from it	(Passarelli et al., 2023)

FIGURE 1 Agriculture framed as innovation receiver/system

Magnitude and Sign at Population Level

Because no study directly models “agriculture alone” as an explanatory variable for **overall innovation in society**, there is:

- **No direct, quantified estimate** of sign (positive/negative) at macro level.
- **No empirical basis** in these papers to propose a normalized effect size from −1.00 to 1.00.

Conclusion

Within the provided research, agriculture is consistently analyzed as a domain **where innovation happens and is needed**, not as a quantified macro-level driver of society-wide innovation. There is **no direct evidence** here to assign a direction or magnitude to agriculture’s standalone influence on overall innovation at the population level. Any broader effects must therefore be regarded as indirect or unquantified based on these studies.

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